

DigiSteel-YOLO

Reference Papers Comprehensive Summary

Literature Review: 11 Reference Papers

Steel Surface Defect Detection using YOLO Architectures

Project: Comprehensive Robustness Study of Lightweight YOLO Detectors

Team: Hazem Elerefy, Youssef Sherif, Mohamed Salah, Moamen Esmat, Mahmoud Hisham

Supervisor: Dr. Tarek Ghoneimy

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1. Executive Summary - Master Comparison Table

ID	Paper	Base	mAP@0.5	Params(M)	FPS	Year	Journal
P01	PSF-YOLO	YOLOv11n	82.2%*	1.82	N/A	2025	Nat. Sci. Reports
P02	LAM-YOLOv10n	YOLOv10n	94.39%	N/A	154	2025	Nat. Sci. Reports
P03	YOLO-LSDI	YOLOv11n	83.0%	2.7	162.1	2025	MDPI Electronics
P04	Lightweight-YOLOv8	YOLOv8n	78.6%	2.04	171.5	2025	Nat. Sci. Reports
P05	SCCI-YOLO	YOLOv8n	78.6%	1.68	270.2	2025	Nat. Sci. Reports
P06	ELS-YOLO	YOLOv11n	79.5%	2.36	N/A	2025	MDPI Electronics
P07	ASFRW-YOLO	YOLOv5s	83.2%	6.20	125	2025	Nat. Sci. Reports
P08	MSFE-YOLO	YOLOv11s	79.8%	11.69	89.3	2026	MDPI Sensors
P09	EFEN-YOLOv8	YOLOv8n	80.4%	N/A	N/A	2026	PLOS ONE
P10	KDM-YOLO	YOLOv10n	95.4%	3.29	155.6	2026	MDPI Sensors
P11	YOLOv11-EMD	YOLOv11	94.9%	N/A	N/A	2025	MDPI Math.

* P01 evaluates on GC10-DET+ (not NEU-DET). P10 trains at native 200px (not 640px upscale).

Novel Modules Summary

Paper	Novel Modules
P01 PSF-YOLO	MDF-Neck, Virtual Fusion Head, Attention Concat, GhostConv
P02 LAM-YOLOv10n	GhostConv, SMA (Spatial Multi-Scale Attention), MFFN
P03 YOLO-LSDI	AMSPPF, DSAM, LDConv, Inner-CIoU
P04 Lightweight-YOLOv8	GhostNet backbone, MPCA, Siou loss
P05 SCCI-YOLO	SPD-Conv, C2f_EMA, CCFM, Inner-IoU
P06 ELS-YOLO	C3k2_THK, Staged-Slim-Neck, MSDetect
P07 ASFRW-YOLO	ASF (SSFF+TFE+CPAM), RepNCSPeLAN4, WIoU v3
P08 MSFE-YOLO	MSFC, C2MSDA, AFfE
P09 EFEN-YOLOv8	SACConv, LSKA, WASPP, gamma-FEIou
P10 KDM-YOLO	KWConv, C2f-DRB, MSAF
P11 YOLOv11-EMD	InnerEIou, MSDA, C3k2_DynamicConv, Transfer Learning

P01 - PSF-YOLO

Full Title

A lightweight YOLOv11-based framework for small steel defect detection with a newly enhanced feature fusion module

Authors

Yongyao Wang, Haiyang Sun, Kai Luo, Quanfu Zhu, Haoifei Li, Yuyang Sun, Zhenjie Wu, Gang Wang

Publication

Scientific Reports (Nature Portfolio), 2025, Vol. 15, Article 34322

DOI: 10.1038/s41598-025-16619-9

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv11n	1.82M	82.2% (GC10-DET+)	45.8% (GC10-DET+)	N/A

Novel Modules

MDF-Neck (Multi-Dimensional-Fusion Neck integrating P1 layer), Virtual Fusion Head (resolution-aware pooling + alignment), Attention Concat Module (spatial-channel attention outperforming SE/CBAM), GhostConv (parameter reduction)

Training Details

Optimizer: SGD; Dataset: GC10-DET+ (enhanced); Split: 7:1:2; Augmentation: flipping, rotation, mosaic

Strengths

1) 25% parameter reduction with +3.2% mAP gain. 2) Strong cross-domain generalization (3 datasets). 3) Principled design with individual ablation.

Weaknesses

1) No FPS/latency benchmarks. 2) No evaluation on NEU-DET. 3) GC10-DET+ dataset not publicly available.

P02 - LAM-YOLOv10n

Full Title

Steel surface defect detection algorithm based on improved YOLOv10

Authors

Laomo Zhang, Zhike Wang, Ying Ma, Guowei Li

Publication

Scientific Reports (Nature Publishing Group), 2025, Vol. 15, Article 32827

DOI: 10.1038/s41598-025-16725-8

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv10n	N/A	94.39%	N/A	154

Novel Modules

GhostConv (lightweight ghost module), SMA (Spatial Multi-Scale Attention - 3 parallel branches with multi-scale pooling and cross-spatial attention), MFFN (Multi-Branch Feature Fusion Network - GAP+conv and local pooling branches)

Training Details

Hardware: GTX 1060Ti; Epochs: 100; Batch: 16; LR: 0.001; Optimizer: Adam; Dataset: NEU-DET (augmented as PRO-DataSet)

Strengths

1) 96.96% precision as nano model (outperforms YOLOv10b). 2) Well-designed modular architecture. 3) Competitive speed (154 FPS).

Weaknesses

1) Missing param count/FLOPs/mAP@0.5:0.95. 2) Only NEU-DET (no cross-dataset). 3) Outdated hardware (GTX 1060Ti).

P03 - YOLO-LSDI

Full Title

YOLO-LSDI: An Enhanced Algorithm for Steel Surface Defect Detection Using a YOLOv11 Network

Authors

Fuqiang Wang, Xinbin Jiang, Yizhou Han, Lei Wu

Publication

Electronics (MDPI), 2025, Vol. 14(13), Article 2576

DOI: 10.3390/electronics14132576

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv11n	2.7M	83.0%	~52-57%	162.1

Novel Modules

AMSPPF (Adaptive Multi-Scale Pooling-Fast), DSAM (Deformable Spatial Attention Module), LDConv (Linear Deformable Convolution), Inner-CIoU loss

Training Details

Cross-dataset validation on NEU-DET, GC10-DET, APSPC PCB datasets

Strengths

1) +5.8% mAP with reduced GFLOPs. 2) Real-time at 162.1 FPS. 3) Cross-dataset validation on 3 datasets.

Weaknesses

1) MDPI Electronics (not top-tier). 2) Incremental contributions. 3) Limited ablation clarity.

P04 - Lightweight-YOLOv8

Full Title

A lightweight algorithm for steel surface defect detection using improved YOLOv8

Authors

Shuangbao Ma, Xin Zhao, Li Wan, Yapeng Zhang, Hongliang Gao

Publication

Scientific Reports (Nature Publishing Group), 2025, Vol. 15, Article 8966

DOI: 10.1038/s41598-025-93469-5

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv8n	2.04M	78.6%	44.5%	171.5

Novel Modules

GhostNet backbone (replaces CSPDarknet), MPCA (Multi-Path Coordinate Attention - max/avg pooling along W and H), Siou loss (angle+distance+shape+IoU costs)

Training Details

GPU: RTX 4090; Epochs: 300; Batch: 16; Optimizer: SGD (lr=0.01, cosine); Image: 640x640; Split: 90/10

Strengths

1) Extreme lightweight (2.04M, 5.1 GFLOPs). 2) ALL metrics improved simultaneously. 3) Edge deployment validated (Raspberry Pi: 115.7 FPS, Jetson: 440.3 FPS). 4) Highly cited (37 citations).

Weaknesses

1) Only NEU-DET. 2) Poor on crazing (21.4% AP). 3) No comparison with 2024 SOTA.

P05 - SCCI-YOLO

Full Title

An efficient and lightweight algorithm for detecting surface defects of steel based on SCCI-YOLO

Authors

Huixiang Zhou, Hong Zou, Gaojun Hu

Publication

Scientific Reports (Nature Portfolio), 2025, Vol. 15, Article 36276

DOI: 10.1038/s41598-025-20154-y

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv8n	1.68M	78.6%	N/A	270.2

Novel Modules

SPD-Conv (Space-to-Depth for small defect detection), C2f_EMA (C2f + EMA attention), CCFM (Cross-scale Convolutional Feature Module), Inner-IoU loss

Training Details

Standard YOLOv8 training protocol on NEU-DET

Strengths

1) Ultra-lightweight (1.68M params, 43.9% reduction). 2) Fastest inference (270.2 FPS). 3) Effective small defect detection via SPD-Conv.

Weaknesses

1) Modest absolute mAP (78.6%). 2) Single dataset (NEU-DET only). 3) Lightweight-accuracy trade-off.

P06 - ELS-YOLO

Full Title

ELS-YOLO: Efficient Lightweight YOLO for Steel Surface Defect Detection

Authors

Zhiheng Zhang, Guoyun Zhong, Peng Ding, Jianfeng He, Jun Zhang, Chongyang Zhu

Publication

Electronics (MDPI), 2025, Vol. 14(19), Article 3877

DOI: 10.3390/electronics14193877

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv11n	2.36M	79.5%	43.2%	N/A

Novel Modules

C3k2_THK (T-shaped conv + Heterogeneous Kernel Selection + SCSA attention), Staged-Slim-Neck (DGSCConv-L/H + GMLCA attention), MSDetect (MRFB multi-scale detection head)

Training Details

GPU: RTX 4070Ti Super; Epochs: 400; Batch: 16; Optimizer: AdamW (lr=0.001); Image: 640x640

Strengths

1) Exceptional param efficiency (2.36M, 5.6G FLOPs). 2) Strong multi-scale detection. 3) Cross-dataset generalization (NEU-DET, GC10-DET, Severstal).

Weaknesses

1) mAP@50:0.95 gap (43.2% vs 44.1% baseline). 2) No FPS reported. 3) Insufficient small-scale defect detection.

P07 - ASFRW-YOLO

Full Title

A high precision and lightweight method for steel surface defect detection based on improved YOLOv5

Authors

Mudan Zhou, Haoyu Wang, Yuhao Wang

Publication

Scientific Reports (Nature Publishing Group), 2025

DOI: 10.1038/s41598-025-28022-5

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv5s	6.20M	83.2%	46.4%	~125

Novel Modules

ASF (Attentional Scale Sequence Fusion: SSFF 3D conv + TFE + CPAM), RepNCSPELAN4 (from YOLOv9, replaces C3), WIoU v3 loss (dynamic non-monotonic focusing)

Training Details

GPU: RTX 4060 Laptop; Epochs: 300; Batch: 16; Optimizer: SGD; Image: 640x640

Strengths

1) Strong mAP improvement (+7.0%). 2) Best mAP@0.5:0.95 (46.4%). 3) Parameter efficiency (fewer than YOLOv5s baseline). 4) Excellent on small defects (crazing +14.4%).

Weaknesses

1) Struggles with crack-type defects. 2) Lab-only evaluation. 3) Limited dataset scope (NEU-DET only).

P08 - MSFE-YOLO

Full Title

MSFE-YOLO: A Steel Surface Defect Detection Algorithm Integrating Multi-Scale Frequency Domain and Defect-Aware Attention

Authors

Siqi Su, Jiale Shen, P. Lin, Wanhe Tang, Weijie Zhang, Zhen Chen

Publication

Sensors (MDPI), 2026, Vol. 26(8), Article 2311

DOI: 10.3390/s26082311

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv11s	11.69M	79.8%	N/A	89.3

Novel Modules

MSFC (Multi-Scale Frequency-Enhanced Conv with Laplacian + depth-adaptive dilation), C2MSDA (Cross-Stage Defect-Aware Attention: Sobel edge + spatial + channel), AFFE (Adaptive Feature Fusion Enhancement)

Training Details

GPU: RTX 3090; Epochs: 300; Batch: 16; Optimizer: SGD; Image: 640x640; GC10-DET: 34.4% mAP@0.5:0.95

Strengths

1) Effective frequency-domain fusion (Laplacian + Sobel). 2) Comprehensive defect-aware attention. 3) Real-time capable (89.3 FPS).

Weaknesses

1) Modest mAP gain (+1.7%) at high cost (+30% params, -35% FPS). 2) Low mAP@0.5:0.95 on GC10-DET. 3) Recall regression (72.6% vs 75.5%).

P09 - EFEN-YOLOv8

Full Title

EFEN-YOLOv8: Surface defect detection network based on spatial feature capture and multi-level weighted attention

Authors

Meishun Wu, Jinmin Peng, Xinyi Yu, Heng Xu, Haotian Sun

Publication

PLOS ONE, 2026

DOI: 10.1371/journal.pone.0339617

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv8n	N/A	80.4%	N/A	N/A

Novel Modules

SACnv (Shallow Attention Conv - multi-scale heterogeneous kernels), LSKA (Large Separable Kernel Attention - cascaded 1D), WASPP (Weighted Atrous Spatial Pyramid Pooling), gamma-FEIou loss (EIou + Focal + adaptive class weighting)

Training Details

GPU: RTX 3060; Epochs: 350; Batch: 16; Optimizer: SGD; 5 random seeds for statistical validation; GC10-DET: 72.1%

Strengths

1) +7.4% mAP with rigorous statistical validation (CIs, t-tests). 2) Modular ablation (WASPP +3.5%, gamma-FEIou +3.0%, LSKA +2.2%). 3) Cross-dataset (NEU-DET + GC10-DET). 4) Open access with code.

Weaknesses

1) Increased computational cost (no FPS). 2) Poor on cracking (44.4%). 3) SACnv marginal alone (+0.2%).

P10 - KDM-YOLO

Full Title

Lightweight Visual Localization of Steel Surface Defects for Autonomous Inspection Robots Based on Improved YOLOv10n

Authors

Jinwu Tong, Xin Zhang, Xinyun Lu, Han Cao, Lengtao Yao, Bingbing Gao

Publication

Sensors (MDPI), 2026, Vol. 26(7), Article 2132

DOI: 10.3390/s26072132

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv10n	3.29M	95.4%	N/A	155.6

Novel Modules

KWConv (KernelWarehouse dynamic kernel sharing), C2f-DRB (Dilated Residual Block with re-parameterization), MSAF (Multi-Scale Attention Fusion: Region + Pixel attention with gated fusion)

Training Details

GPU: RTX 4060; Epochs: 200; Batch: 16; Optimizer: SGD; Image: 200x200 (native); Cross-domain: bearing defect +3.2%

Strengths

1) Highest mAP (95.4%) with lightweight design (3.29M, 155.6 FPS). 2) Excellent ablation (cumulative gains). 3) Cross-domain generalization. 4) Cracking: +24.8% AP improvement.

Weaknesses

1) No mAP@0.5:0.95 reported. 2) Trained at native 200px (protocol difference). 3) Small dataset (1,800 images).

P11 - YOLOv11-EMD

Full Title

YOLOv11-EMD: An Enhanced Object Detection Algorithm Assisted by Multi-Stage Transfer Learning for Industrial Steel Surface Defect Detection

Authors

Weipeng Shi, Junlin Dai, Changhe Li, Na Niu

Publication

Mathematics (MDPI), 2025, Vol. 13(17), Article 2769

DOI: 10.3390/math13172769

Base Model & Metrics

Base Model	Params	mAP@0.5	mAP@0.5:0.95	FPS
YOLOv11	N/A	94.9%	N/A	N/A

Novel Modules

InnerEIoU loss (improved EIoU), MSDA (Multi-Scale Dilated Attention), C3k2_DynamicConv (attention-weighted kernel combinations), Multi-stage Transfer Learning (source pre-train + target fine-tune)

Training Details

Multi-stage transfer learning; Datasets: NEU-DET + Severstal (combined), NEU-DET + GC10-DET (cross-scenario); Transfer learning: -3.2% time, +8.8% mAP

Strengths

1) Practical transfer learning framework (+8.8% mAP). 2) Multi-dataset evaluation. 3) Targeted improvements for specific challenges.

Weaknesses

1) Limited metrics (no FPS, params, mAP@0.5:0.95). 2) Short paper (6 pages). 3) Cross-scenario gap (94.9% to 79.9%).

13. Comparative Analysis

Accuracy vs. Efficiency Trade-off

Paper	mAP@0.5	Params(M)	FPS	Rating
P10 KDM-YOLO	95.4%	3.29	155.6	Best accuracy-efficiency
P02 LAM-YOLOv10n	94.39%	N/A	154	High accuracy
P11 YOLOv11-EMD	94.9%	N/A	N/A	Transfer learning
P07 ASFRW-YOLO	83.2%	6.20	125	Best localization
P03 YOLO-LSDI	83.0%	2.7	162.1	Balanced
P09 EFEN-YOLOv8	80.4%	N/A	N/A	Statistical rigor
P08 MSFE-YOLO	79.8%	11.69	89.3	Frequency domain
P06 ELS-YOLO	79.5%	2.36	N/A	Most efficient
P04 Lightweight-YOLOv8	78.6%	2.04	171.5	Edge deployment
P05 SCCI-YOLO	78.6%	1.68	270.2	Fastest inference

Key Observations for DigiSteel-YOLO

1. NO paper reports robustness evaluation - DigiSteel's perturbation framework (6 types x 4 levels) is unique.
2. Training protocol inconsistency - P10 trains at 200px native; others at 640px. Direct comparison is unfair.
3. GhostConv is popular (P01, P02, P04) - validates our use in DigiSteel v2.
4. Inner-IoU variants trending (P03, P05, P11) - validates our Inner-WIoU choice.
5. EMA attention used by P05 - same as DigiSteel v2. Common technique, not novel.
6. Best achievable on NEU-DET: ~95% (P10 at 200px). At 640px: ~83% is competitive range.
7. Most papers use NEU-DET (10/11) and SGD optimizer with 200-300 epochs.
8. Lightweight models (P04, P05) achieve 78.6% with <2.1M params - efficiency frontier.